Model	Semantic Entropy		Probability	
	Random	CHOKE	Random	CHOKE
Llama	5.2±1.8	**15.8**±6.6	5.1±1.4	**25.7**±12.8
Mistral	6.4±1.0	**18.2**±3.9	13.6±2.5	**40.6**±12.1
Gemma	4.3±1.3	**13.0**±5.5	5.7±1.5	**27.8**±15.5
Llama-Inst	5.3±1.4	**18.0**±7.0	8.6±2.2	**25.8**±11.5
Mistral-Inst	10.2±2.4	**24.9**±9.2	9.3±2.1	**31.2**±13.4
Gemma-Inst	7.7±2.4	**26.7**±12.5	9.2±2.1	**31.7**±16.6

Table 2: **CHOKE examples are more consistent across prompts than other hallucinations.** The *CHOKE* columns show high Jaccard similarity across prompts, indicating strong consistency, while randomly sampled hallucinations (*Random*) have low similarity. All results are statistically significant (permutation test, $p < 0.008$).

hallucinations are comparable to those observed in Gemma-9B. This suggests that this phenomenon also exist in larger models. See Appendix H for similar results on the TriviaQA dataset.

4.3 CHOKE Examples Cannot Be Explained as Noise

While the existence of CHOKE examples is apparent, a potential criticism is that these samples could merely reflect noise stemming from the natural correlation between uncertainty and hallucinations, rather than constituting a distinct and consistent subset. To address this, we evaluate the similarity of CHOKE examples across any two of our prompt variants.

Hallucination and non-hallucination classifications differ significantly between these settings, with overall Jaccard similarity between their hallucinations ranging from 30% to 50%. Thus, finding consistent CHOKE examples across these diverse settings will suggest they are not artifacts of the uncertainty-hallucination correlation but instead represent a robust phenomenon.

We quantify this consistency using the Jaccard similarity of CHOKE examples across settings and validate its uniqueness with a permutation test on 10K randomly sampled subsets of hallucination samples of equivalent size.[3] The results confirm that CHOKE examples similarity between context settings exceeds random expectations, as shown in Table 2, using semantic entropy and probability metrics. Appendix G provides additional analyses, including results for TriviaQA and for only shared hallucination examples between the settings that further demonstrate the uniqueness of CHOKE.

[3]We ran this for any two settings and report the mean.

5 CHOKE-Score

Having established the existence and distinctiveness of CHOKE examples, we hypothesize that the performance of hallucination mitigation methods will differ when evaluated on CHOKE examples, compared to their overall effectiveness. These cases represent a unique challenge, and assessing mitigation on them may expose limitations or trade-offs not captured by standard metrics. To this end, we introduce *CHOKE-Score*, a novel evaluation metric for assessing the ability of hallucination mitigation methods to reduce CHOKE examples. This perspective is particularly valuable in high-stakes settings, where model certainty impacts decisions.

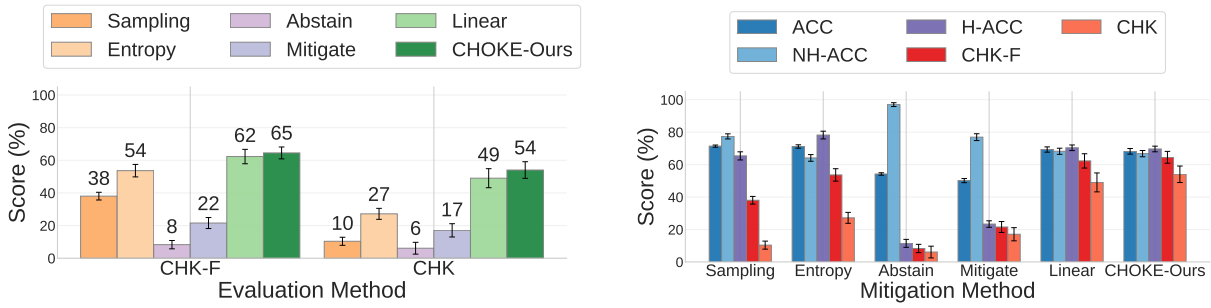
Formally, given CHOKE examples detected via a certainty measurement method d (Section 3.2), we measures the proportion successfully mitigated:

$$\text{CHK-d}_{(\mathcal{M},\mathcal{C}_d)} = \frac{|\mathcal{M} \cap \mathcal{C}_d|}{|\mathcal{C}_d|} \tag{2}$$

Here, $\mathcal{C}_d$ is the set of CHOKE examples flagged by a detection method d, and $\mathcal{M}$ is the set of successfully mitigated hallucinations. The score ranges from 0 to 1, with 1 indicating that all $\mathcal{C}_d$ examples were mitigated (found in $\mathcal{M}$). To ensure a broad definition of CHOKE-Score, we combine all three detection methods from Section 3.2 to create two score variations:

$$\text{CHK}_{(\mathcal{M},\mathcal{C}_\cap)} = \frac{|\mathcal{M} \cap \mathcal{C}_\cap|}{|\mathcal{C}_\cap|} \tag{3}$$

$$\text{CHK-F}_{(\mathcal{M},\mathcal{C}_\cup)} = \frac{|\mathcal{M} \cap \mathcal{C}_\cup|}{|\mathcal{C}_\cup|} \tag{4}$$



(a) CHOKE-Score for different mitigation methods.

(b) CHOKE-Score vs. standard metrics.

Figure 5: **Our mitigation outperforms other methods on CHOKE-Score and CHOKE-Score reveals limits of standard methods.** Figure 5a (left) shows our probe method achieves the highest CHOKE-Score scores. Figure 5b (right) compares CHOKE-Score (red) to other metrics (blue shades), showing that certainty-based methods perform well generally but poorly on CHOKE-Score, exposing gaps in handling CHOKE examples. Probe methods maintain more consistent performance, demonstrating stronger robustness. Scores averaged over six models and all prompts.

where $\mathcal{C}_\cap$ is the set of CHOKE examples flagged by all detection methods (*intersection*), and $\mathcal{C}_\cup$ is the set flagged by at least one method (*union*). These two variants allow us to evaluate mitigation effectiveness under strict (CHK) and flexible (CHK-F) detection criteria.

While CHK targets strict cases flagged by all methods, CHK-F includes any case flagged by at least one method. Together, they offer a fuller picture of mitigation performance.

5.1 Mitigation Methods

We use the CHOKE-Score to evaluate three hallucination mitigation approaches: certainty-based, prompt-based, and probe-based, including our own variant: CHOKE-tuned probe method.[4]

5.1.1 Baseline Methods

Certainty methods. Certainty-based mitigation leverages uncertainty estimates to determine when to abstain from generation. Such approaches rely on self-evaluation techniques (Tomani et al., 2024), information-theoretic measures (Yadkori et al., 2024a,b), or use cross-verification across multiple models to assess uncertainty (Feng et al., 2024). We employ a **sampling**-based method following Cole et al. (2023) and a **predictive entropy**-based method following Tomani et al. (2024).

Prompting methods. Prompt-based methods aim to improve factuality or guide the model toward abstention or generating truthful responses by using crafted prompts (Feng et al., 2024; Taveekitworachai et al., 2024). Specifically, we use a self-reflect prompt (Feng et al., 2024) and null-shot prompting (Taveekitworachai et al., 2024). As these methods require instructions, we evaluate them only on instruction-tuned models.

Probing methods. Probe-based methods use classifiers trained on internal activations to detect and abstain from hallucinations (Li et al., 2023; CH-Wang et al., 2023). We apply logistic regression on residual stream activations at the last token of the 15th layer, following CH-Wang et al. (2023).

5.1.2 CHOKE-tuned Mitigation (Ours)

We augment the standard linear probe by oversampling CHOKE examples during training, specifically increasing their proportion in training to 65%. We hypothesize this will boost CHOKE-Score performance with minimal impact on overall accuracy.

5.2 Results

We report both CHOKE-Score variants—CHK (strict) and CHK-F (flexible)—for each mitigation. For context, we include overall accuracy (ACC), hallucination accuracy (H-ACC), and non-hallucination accuracy (NH-ACC). Figure 5 shows the results averaged across six models.[5]

CHOKE examples challenge mitigation. Figure 5a compares the performance of the different mitigation methods on the CHOKE-Score. Prompting methods perform worst, followed by certainty-based mitigation. Probe-based methods perform better, and our CHOKE-Tuned performs best. This demonstrates that focusing on CHOKE examples

[4]When required, training uses a 50%/50% random split.

[5]Results are on NaturalQA. Similar results on TriviaQA and additional metrics are in Appendix I.

may help improve on CHOKE-Score, an important consideration in domains where reliable, high-certainty predictions are essential. However, while the score increased, fully mitigating high-certainty hallucinations despite having the correct knowledge remains challenging.

CHOKE-Score reveals a hidden performance gap. Figure 5b presents a clear trend: methods strong on general metrics often underperform on CHOKE. While certainty-based methods achieve high accuracy overall and on general hallucinations, their CHOKE-Score is much lower. Prompt-based methods perform well on some metrics yet show the largest drop in CHOKE-Score. In contrast, linear probe mitigation remains stable across all metrics, with only a minor gap on CHOKE-Score.

These findings support our hypothesis that CHOKE examples differ from general hallucinations and require separate evaluation. Traditional metrics often overlook this, hiding methods' weaknesses on CHOKE examples. CHOKE-Score fills this gap, offering a more accurate picture of a method's robustness, especially where model certainty matters. Our CHOKE-tuned mitigation boosts performance, showing that targeted training can enhance robustness on CHOKE examples.

6 Discussion and Conclusion

This work investigated high certainty hallucinations occurring despite the model having the knowledge to answer correctly—a phenomenon we termed CHOKE. While hallucinations are typically linked to uncertainty or ignorance, CHOKE, arises with both certainty and sufficient knowledge. CHOKE examples are especially concerning in high-stakes domains, where certainty often proxies reliability.

To address this, we introduce CHOKE-Score, a new metric for evaluating mitigation methods on CHOKE. While existing mitigation methods perform well on standard benchmarks, they struggle on CHOKE-Score. To overcome this, we propose a new method that outperforms others on this metric. Our findings highlight the need to understand CHOKE and develop targeted mitigation.

7 Limitations

This work demonstrates the existence of CHOKE and shows that it presents a significant challenge for detection and mitigation methods. However, this work does not offer an explanation for why this phenomenon occurs or what triggers it. Further research is needed to deepen our understanding of the underlying causes. Moreover, the proposed mitigation solution, while improving performance on the CHOKE-Score, remains far from optimal. Lastly, the work did not address other types of hallucinations (e.g., lack of knowledge), nor did it introduce evaluation mitigation metrics tailored to those types.

8 Ethic Statement

In this work, we demonstrated the existence of CHOKE, proposed a pipeline to detect it, and explored ways to mitigate it. While both the phenomenon and its mitigation could potentially be misused to make models less reliable—yet appear certain—our goal is to advance the understanding of the CHOKE phenomenon to ultimately enhance model reliability. We also conducted a human annotation study to assess prompt neutrality. Participation was voluntary and anonymous, with no personal or sensitive data collected. Annotators gave informed consent for their anonymized responses to be used in the study. As the task posed minimal risk and involved no sensitive content, it was deemed exempt from formal ethics review.

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